

TASK MANAGEMENT WEB APPLICATION

‘TASK_FLOW’



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Date Of Submission

2026

PROJECT PROPOSAL

TASK_FLOW – TASK MANAGEMENT WEB APPLICATION

1. INTRODUCTION

1.1 Background of the Study

Task management plays a vital role in improving productivity in both academic and professional environments. Students and employees often struggle to manage daily tasks, deadlines, and priorities effectively. Traditional task management methods such as handwritten notes or basic reminder apps lack structured organization, search capability, and progress tracking features.

With the advancement of web technologies, modern browsers allow applications to store data locally without requiring a database or backend server. This project focuses on developing a simple web-based task management system using only front-end technologies.

1.2 Problem Statement

Many individuals use manual or unstructured systems to manage their daily tasks. These systems:

- Do not provide filtering or sorting features
- Do not show task progress statistics
- Make it difficult to track overdue tasks
- Lack digital organization

There is a need for a simple, user-friendly web application that enables efficient task management without complex server setup or database configuration.

1.3 Aim of the Project

The aim of this project is to design and develop a single-user task management web application using HTML, CSS, and JavaScript, where all task data is stored using browser LocalStorage.

1.4 Objectives

The objectives of this project are:

1. To design a responsive and user-friendly interface.
 2. To implement task creation, editing, and deletion (CRUD operations).
 3. To allow users to set task priority and due dates.
 4. To implement task status tracking (To Do, In Progress, Done).
 5. To provide search, filtering, and sorting features.
 6. To display dashboard statistics (Total, Completed, Overdue tasks).
 7. To store and retrieve data using LocalStorage.
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2. LITERATURE REVIEW

Task management systems have evolved significantly with modern web technologies. Applications such as:

- Todoist
- Microsoft To Do
- Google Tasks
- Trello

provide task management features using cloud-based storage systems. However, these systems require:

- Internet connectivity
- Server infrastructure
- User authentication
- Database management

Recent web technologies provide alternatives such as the Web Storage API, which allows client-side data storage. LocalStorage enables developers to store structured JSON data directly in the browser. This eliminates the need for backend infrastructure while still allowing persistent data storage.

The proposed system uses LocalStorage as the primary data management mechanism.

3. PROPOSED SYSTEM

The proposed system is a browser-based single-user task management application.

Key Characteristics:

- No login system
- No database
- No backend server
- Fully client-side architecture
- Offline functionality

All tasks are stored locally in the browser using JSON format.

4. FUNCTIONAL REQUIREMENTS

The system shall:

1. Allow users to add new tasks.
 2. Allow users to edit existing tasks.
 3. Allow users to delete tasks.
 4. Allow users to set priority (Low, Medium, High).
 5. Allow users to assign due dates.
 6. Allow users to update task status.
 7. Display dashboard statistics.
 8. Allow search by task title.
 9. Allow filtering by task status.
 10. Allow sorting by date and priority.
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5. NON-FUNCTIONAL REQUIREMENTS

1. The system must be responsive and user-friendly.
 2. The system must load quickly.
 3. The system must work offline.
 4. Data must persist after page refresh.
 5. The system should be easy to maintain and extend.
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6. MATERIALS AND TECHNOLOGIES

6.1 Hardware Requirements

- Computer or laptop
- Minimum 4GB RAM
- Modern web browser

6.2 Software Requirements

- Operating System (Windows/Linux)
- Visual Studio Code or Notepad++
- Web Browser (Chrome, Firefox, Edge)

6.3 Technologies Used

- HTML – For structure
 - CSS – For styling and layout
 - JavaScript – For functionality
 - LocalStorage – For data storage
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7. SYSTEM ARCHITECTURE

The system follows a client-side architecture:

- User Interface Layer (HTML + CSS)
- Logic Layer (JavaScript)
- Storage Layer (LocalStorage)

There is no server or database involved. All operations are performed inside the browser.

8. METHODOLOGY

The project follows the Software Development Life Cycle (SDLC):

1. Requirement Analysis
 2. System Design
 3. Implementation
 4. Testing
 5. Deployment
 6. Documentation
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9. TIME PLAN

Phase	Duration
Requirement Analysis	Week 1
Interface Design	Week 2
Development	Week 3
Testing	Week 4
Final Report Preparation	Week 5

10. EXPECTED OUTCOME

The expected outcome of this project is a fully functional single-user task management web application that:

- Runs in any modern browser
 - Works offline
 - Stores data permanently in LocalStorage
 - Provides organized task management features
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11. CONCLUSION

This project demonstrates how a practical web application can be developed using only front-end technologies without backend integration. The system provides an efficient solution for managing daily tasks while maintaining simplicity and ease of use.

12. RISK ANALYSIS

Risk analysis is an important part of project planning. It helps identify potential problems that may occur during development and implementation. The following risks have been identified for the TaskFlow project.

12.1 Technical Risks

1. Browser Compatibility Issues

Different browsers may behave slightly differently when handling LocalStorage or JavaScript functions.

Mitigation:

The system will be tested on major browsers such as Google Chrome, Mozilla Firefox, and Microsoft Edge.

2. Data Loss Risk

Since the application uses LocalStorage, task data may be lost if:

- The user clears browser data
- The browser is reset
- The system is used on a different device

Mitigation:

Users will be informed that data is stored locally. Future improvements may include export/import functionality.

3. Code Errors / Bugs

Logical errors or incorrect implementation may cause malfunctioning features.

Mitigation:

- Unit testing will be performed for each module.
 - Debugging tools in browser developer console will be used.
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12.2 Project Management Risks

1. Time Management Risk

Delays in development may occur due to academic workload or unforeseen issues.

Mitigation:

A proper schedule and weekly milestones will be followed.

2. Requirement Misunderstanding

Incorrect interpretation of project requirements may lead to rework.

Mitigation:

Requirements will be clearly documented and reviewed before development.

12.3 Operational Risks

1. User Misuse

Users may accidentally delete all tasks.

Mitigation:

Confirmation dialogs are included before deleting tasks or clearing all tasks.

12.4 Risk Summary Table

Risk	Impact	Probability	Mitigation
Data Loss	High	Medium	User awareness + future backup feature
Browser Issues	Medium	Low	Cross-browser testing
Development Delay	Medium	Medium	Follow project schedule
Software Bugs	Medium	Medium	Testing and debugging

13. GANTT CHART EXPLANATION

A Gantt chart is used to visually represent the project schedule. It shows the start and end dates of each activity and helps monitor project progress.

For the TaskFlow project, the development process is divided into several phases.

13.1 Project Phases

Phase 1 – Requirement Analysis (Week 1)

- Identify system requirements
 - Define project objectives
 - Prepare proposal
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Phase 2 – System Design (Week 2)

- Design user interface layout
 - Plan data structure
 - Define system architecture
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Phase 3 – Development (Week 3 – Week 4)

- Implement HTML structure
 - Design CSS styling
 - Implement JavaScript logic
 - Integrate LocalStorage
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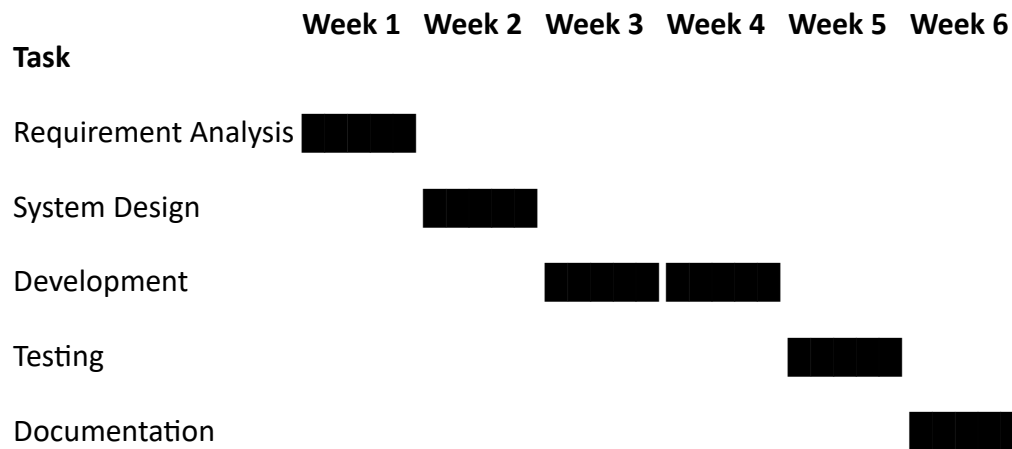
Phase 4 – Testing (Week 5)

- Perform functional testing
 - Debug errors
 - Validate features
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Phase 5 – Documentation and Finalization (Week 6)

- Prepare final report
 - Prepare presentation slides
 - Final review and submission
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13.2 Gantt Chart Timeline Overview



The Gantt chart helps ensure that:

- The project stays on schedule
- Each phase is completed before moving to the next
- Deadlines are clearly defined
- Progress can be tracked easily